

49FlashMoney Product Requirements Document

v1.0

Volume 4 – Technical Architecture, Security, DevOps & QA

System Architecture

Frontend: React + TypeScript + Vite. Backend: Django, DRF, Django Channels, Celery. Data: PostgreSQL, Redis. Storage: Object storage. Edge: CDN and reverse proxy.

Database

Use an immutable wallet ledger. Separate operational tables from reporting. Apply foreign keys, indexes, audit fields, and transactional integrity for all money movements.

API Standards

REST APIs with versioning, JWT authentication, request validation, pagination, idempotency for payment endpoints, consistent error model, and OpenAPI documentation.

WebSocket Protocol

Support live game rounds, countdown timers, live bets, jackpots, notifications, and reconnect semantics with authenticated channels.

Security

KYC integration, MFA/OTP, RBAC, rate limiting, CSRF/XSS protection, input validation, encryption in transit and at rest, audit logs, fraud detection, and secret management.

DevOps

Dockerized services, CI/CD, automated tests, staging and production environments, infrastructure as code, blue/green deployment, backups, rollback procedures.

Observability

Centralized logs, metrics, tracing, dashboards, alerting, health checks, SLA monitoring, and business KPI dashboards.

Testing

Unit, integration, API, UI, load, security, regression, and end-to-end tests. Critical financial workflows require automated coverage before release.

Production Readiness Checklist

Disaster recovery tested, monitoring enabled, backups verified, payment reconciliation complete, feature flags configured, security review passed, documentation updated.